ISYS2101 – Software Engineering Project Management

Project Proposal and Planning

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I declare that in submitting all work for this assessment I have read, understood and agreed to the content and expectations of the Assessment declaration.

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Executive Summary

This document presents GreenRoute HCMC, a proposal for an EV-focused route guidance system available on both Android and the web. It outlines the project rationale, objectives, scope, technology choices, product features, and the criteria that will be used to evaluate whether the project has been successfully completed. The proposed system focuses on practical EV travel by considering route feasibility, battery range, charging accessibility, and local air-quality conditions rather than relying only on the shortest or fastest path.

The project is timely because Vietnam has committed to achieving net-zero emissions by 2050, has entered the Just Energy Transition Partnership with the EU and other partners, and is continuing to promote greener transport policies. At the city level, Ho Chi Minh City is already pursuing electric and green-energy transport through an electrification roadmap for buses and motorbike-based services, making an EV-oriented navigation solution locally relevant rather than theoretical [1] [2].

The proposed MVP is realistic because it focuses on a pilot area in HCMC and mainly supports electric motorbike or scooter journeys, while keeping the system architecture extensible for future interprovincial travel. Rather than attempting to build a full commercial navigation ecosystem from scratch, the project combines existing routing and air-quality services with its own battery-feasibility logic and forward-progress charging-stop recommendation. This allows the system to guide users not only toward reachable charging points, but toward ones that continue the journey efficiently rather than causing unnecessary detours. The system is also designed with a privacy-conscious approach, where personal trip data can remain on the user's device rather than being centrally stored.

1. Problem statement

1.1. Project purpose

Current navigation apps are good at answering one question: Which route is fastest? For EV users, however, that is not enough. An EV rider also needs to know whether the destination is reachable with the remaining battery, how far the vehicle can still travel based on its model and current charge, and where the most sensible charging or battery-swap stop is when the battery is insufficient. In practice, a route is only useful if it can actually be completed with the energy available.

This problem is especially relevant in Ho Chi Minh City, where electric two-wheelers are becoming more common and where commuting and delivery trips are often dense, stop-and-go, and time-sensitive. A rider does not just need "a route"; they need a route that keeps the trip viable from start to destination. That makes battery-aware routing a meaningful problem for both daily commuters and last-mile delivery users. The need is further strengthened by HCMC's green-transport roadmap, which treats air pollution as urgent and targets a major shift toward electric and green-energy vehicles.

The practicality of this problem can already be seen in features that exist across different products and devices. Google Maps can provide route guidance and show nearby charging-station locations, IQAir offers live air-quality maps, and many EVs already display battery level and estimated remaining range through the vehicle interface or companion apps. However, these functions are still separated across multiple platforms, which means users often have to switch back and forth between navigation, charging, air-quality, and vehicle information before deciding how to travel. GreenRoute addresses this inconvenience by bringing these decisions into a single system that helps users choose a route, estimate whether their remaining battery is sufficient, and identify a reachable charging or battery-swap point that still supports continued progress toward the destination. At the same time, the system is not limited to EV riders, as non-EV users can also benefit from the Lower-Pollution Route option when they want a more environmentally aware journey rather than simply the fastest one.

1.2. Project objectives

The objectives of the project are:

1. To develop an Android and web-based EV route guidance system for a pilot area in Ho Chi Minh City, while designing the backend so it can later be extended to interprovincial trips.
2. To estimate the remaining travel distance of an EV based on the user's current battery percentage and selected vehicle model, and to classify trip feasibility into simple confidence levels such as directly reachable, near limit, or requiring a charging stop.
3. To generate three user-selectable route choices for each trip: Optimal Route, Alternative Route, and Lower-Pollution Route.
4. To recommend a reachable charging or battery-swap point that not only matches the vehicle's remaining travel range, but also supports continued progress toward the destination, before recalculating the onward route after charging.
5. To display AQI information by area and incorporate AQI-based weighting into route scoring so that more polluted segments carry a slightly higher effective travel cost without causing impractical detours.
6. To maintain a trip log containing the selected route type, battery sufficiency result, charging-stop recommendation, and completed trip summary, while keeping personal trip data stored locally on the user's device or browser

1.3. Project scope

The project will aim to deliver a cross-platform EV route guidance system with built-in air-quality awareness. The system will be delivered through two platforms:

- A web application, accessible through standard web browsers
- An Android mobile application, designed for smartphones and tablets

Both platforms will allow users to search for routes to their intended destination while taking into account the remaining battery level, vehicle model, reachable charging or battery-swap stations, and pollution conditions along the route. The system will provide three route options: Optimal Route, Alternative Route, and Lower-Pollution Route. If the battery is not sufficient for direct travel, the system will recommend a reachable charging stop within a forward travel corridor toward the destination, rather than simply selecting the closest station by distance. The system will also classify trip feasibility into simple levels such as directly reachable, near limit, or requiring a charging stop. The MVP will focus on a pilot area in Ho Chi Minh City, with the system architecture designed to support future expansion beyond the city.

Included within the project's scope:

- Web-based and Android platforms for EV route guidance and map-based visualization.
- Route planning based on destination, vehicle model, and current battery percentage.
- Vehicle range estimation and battery-feasibility checking.
- Recommendation of reachable charging or battery-swap stations within a forward travel corridor toward the destination.
- Route comparison through:
    - Optimal Route
    - Alternative Route
    - Lower-Pollution Route
- AQI map overlay by district or zone using external air-quality data.
- Trip history/log for route choice, battery feasibility, and charging-stop guidance, stored locally on the user's device or browser.
- Integration with external routing, geocoding, map, and AQI services.

Excluded from the project's scope:

- Full commercial-grade turn-by-turn navigation comparable to Google Maps or Waze
- Real-time charger booking, payment, or occupancy guarantees
- Direct integration with vehicle hardware, telematics, or battery-management systems
- High-accuracy street-level pollution modelling or pollution forecasting
- Advanced live traffic prediction
- Native iOS application
- Nationwide deployment in the MVP stage
- Development of a custom routing engine from scratch

1.4. Product features

1. Battery-aware route guidance

Users enter origin, destination, current battery percentage, and vehicle model. The system estimates whether the destination is directly reachable before finalising route guidance.

2. Three route choices

The system presents:

- **Optimal Route** – the fastest practical route
- **Alternative Route** – another viable route option for user comparison
- **Lower-Pollution Route** – a route with moderate AQI-based penalties applied to more polluted areas, provided the detour stays within a practical threshold

This design keeps the pollution-aware option useful, but prevents it from overriding battery feasibility and overall travel practicality.

3. Vehicle model and reachable-distance estimation

Users select a supported EV model, and the system estimates how far the vehicle can still travel using the current battery percentage and that model's configured full-range assumption. To make the result easier to interpret, the system also presents trip feasibility in simple confidence levels such as directly reachable, near limit, or requiring a charging stop.

4. Forward-direction charging or battery-swap recommendation

If the destination is not directly reachable, the system identifies charging or battery-swap points within the vehicle's current reachable range and ranks them based on how well they support continued travel toward the destination. This allows the system to avoid recommending stations that are technically nearby but require unnecessary backtracking or side detours. After a stop is selected, the system recalculates the next stage of the route from that station to the final destination.

5. AQI overlay and pollution-aware scoring

The map displays AQI by district or selected zone, and that AQI data contributes to the Lower-Pollution Route calculation. Because IQAir's HCMC data includes non-government contributors, the result should be presented as a lower estimated exposure option, not as a guarantee of safe air quality.

6. Trip log and travel summary

The system records previous trip plans and completed journeys, including route type selected, battery sufficiency result, recommended charging stop, and completed trip summary. To support a privacy-conscious design, this information is stored locally on the user's device or browser rather than on a central server.

7. Planned enhancement features

If time permits, the project may include several stretch features that help the app stand out while still staying relevant to navigation:

- **Weather-aware range adjustment**, especially temperature-based range changes and optional wind or elevation effects.
- **Live charger availability or charger filters**, such as plug type, charging speed, or preferred network.
- **Offline route access** for weak-connectivity situations.
  Safety alerts, such as warnings for crash-prone road segments or route hazards.
- **Lane guidance** near complex intersections and major turns.

- **Destination-side convenience features**, such as nearby parking, charger amenities, or rest reminders for longer trips.

1.5. Project success metrics

The project will be considered successful if it meets the following criteria:

Schedule

- The project is completed within the approved capstone timeline.
- Major milestones for analysis, design, implementation, testing, and demonstration are completed on schedule.

Scope

- The final system includes all defined MVP features:
    - vehicle-model-based range estimation,
    - three route choices,
      charging or battery-swap recommendation,
    - AQI overlay,
    - trip log.

Quality and reliability

- All core user flows pass planned functional tests.
- No unresolved critical defects remain at final testing.
- Battery-feasibility classification works correctly in benchmark scenarios covering directly reachable trips, near-limit trips, and trips that require a charging stop.
- In benchmark test scenarios where the destination is not directly reachable, the system recommends a charging or battery-swap point that is both reachable and located within the defined forward travel corridor toward the destination.

Performance

- The system returns route guidance results within an acceptable time, such as **5 seconds or less** under normal demo conditions in the pilot zone.
- Core map interactions and trip-planning screens operate smoothly on a standard Android device.

User effectiveness and satisfaction

- At least 80% of users can successfully complete key tasks such as planning a route, checking battery feasibility, and following a recommended charging stop without assistance.
- User feedback indicates that the system is understandable, practical, and useful for EV travel planning.

- Supervisors and testers agree that the final system clearly demonstrates a realistic solution to the stated problem.
- Users are able to understand the trip-feasibility result and distinguish between directly reachable trips, near-limit trips, and trips requiring a charging stop.

2. Project Plan

2.1. Resource list

2.1.1 Human Resources

The following table summarizes the allocation of human resources throughout the project.

Number of Team Members: 4		
Time allocation	**1 Member**	**Entire team**
Per week	10 hours	40 hours
Entire Project	80 hours	320 hours

2.1.2 Hardware Resources

The hardware resources required for the development and testing of the system are outlined below.

1. **PCs/Laptops:** Used for software development and testing
2. **Android device/Emulator**: Used for Android application testing
3. **Internet Connection:** Required for accessing external APIs and supporting development tools

2.1.3 Software Resources

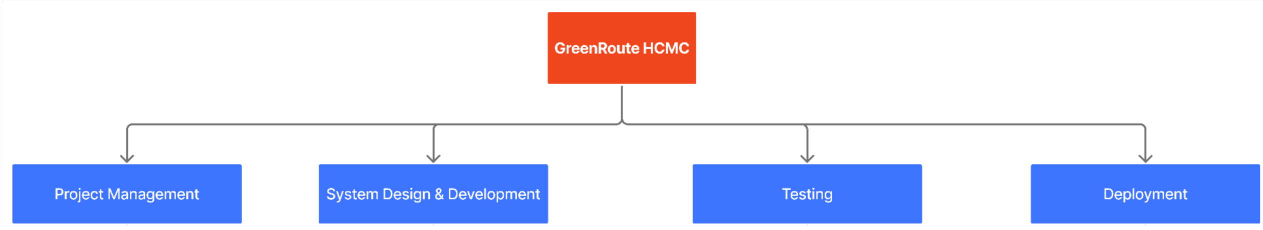
For the frontend of the application, JavaScript, HTML, and CSS will be used as the primary development technologies, with Leaflet [3] handling interactive map rendering and overlays. The application will communicate with the backend through HTTP requests using the Fetch API. For client-side data caching and trip history, browser LocalStorage will be utilized rather than a centralized database.

On the backend, ExpressJS running on Node.js will be used to implement RESTful endpoints that process and serve data related to routing, pollution, and weather. These endpoints will retrieve external data using HTTP requests and return structured responses in JSON format.
For routing functionality, OSRM [4] will be used in conjunction with map data from OpenStreetMap [5]. Environmental data will be integrated using IQAir [6] for air quality information and Open-Meteo [7] for weather conditions such as temperature, humidity, wind, and rainfall. Location-based queries and map feature extraction will be supported through Overpass API [8] and Nominatim [9].

The system will be deployed as a web application and wrapped into a mobile application using Android WebView.

For a comprehensive list of software resources, refer to Appendix A.

2.2. Work breakdown structure



For full details of the work breakdown structure, refer to Appendix B.

2.3. Project schedule

The project will be executed over 8 weeks, with a total team capacity of 320 hours (40 hours/week). Work is structured into sequential phases: setup, core development, integration, testing, and closure, with parallel frontend and backend development where appropriate. Project management activities, including progress tracking, team coordination, and scope management, are conducted continuously throughout the project lifecycle by the Project Manager. Scheduled meetings will be held at the start of each week to determine the specific backlogs for that development week. Meetings during a development week will be scheduled on the go and will be determined by the project manager. We will be using the Parallel Development method [10] to tackle this project. The project will be divided into smaller sub-sections to be tackled simultaneously. This will help reduce the down time during the development process.
For a detailed look of the project schedule, refer to Appendix C.

PROJECT SCHEDULE

Week 1 – Environment Setup & Prerequisites
- **Small Milestone 1:** Development environment fully operational

Week 2 – System Skeleton Development
- **Small Milestone 2**: Working system prototype with UI skeleton

Week 3 – Core Logic Implementation
- **Small Milestone 3:** Fully functional routing system

Week 4 – Data Integration & Mid-Project Review
- **Large Milestone 1:** Real data integration completed and validated

Week 5 – Environmental Overlays
- **Small Milestone 4:** Advanced map features completed

Week 6 – UI Completion & Mobile Integration
- **Small Milestone 5:** Complete user interface and mobile compatibility achieved

Week 7 – Testing & Deployment
- **Large Milestone 2:** System successfully deployed

Week 8 – Project Closure & Documentation
- **Small Milestone 6:** Final project submission and sign-off

2.4. Risk management

To ensure the successful and timely delivery of this project, we conducted a comprehensive risk assessment covering technical, operational, and team-related challenges. The risks fall into 4 critical areas of concern. Isolating these core themes allows us to prioritize our mitigation efforts on the highest-impact threats - specifically those most likely to derail our schedule, compromise the MVP, or disrupt system integration.

- **Schedule & Effort Miscalculation:** Development timelines are tight. Underestimating the time needed for complex modules (like EV range calculation) or failing to stick to a strict MVP could prevent us from delivering on time.
- **Scope Creep:** There is a high likelihood of the project getting derailed by "nice-to-have" features (e.g., weather, offline mode, safety alerts) before the core functionality is fully stable.
- **Technical & Integration Bottlenecks:** Connecting the Android app, web app, backend, and external APIs poses a significant challenge. This is compounded by the risk of inaccurate routing logic or incomplete map/charging station data.
- **Resource & Coordination Gaps:** The project is vulnerable to team-related disruptions, such as uneven workloads, siloed knowledge, or unexpected member absences, which could stall critical development phases.

Complete list of all assessed risks is in Appendix D.

2.5. Assumptions and constraints

Assumptions

The project is developed based on the following assumptions:

1. **Users are able to provide reasonably accurate input data.**
   It is assumed that users can correctly enter or select their origin, destination, current battery percentage, and vehicle model. The accuracy of route feasibility depends partly on this input.
2. **Vehicle model information is available in a usable form.**
   The project assumes that nominal travel-range information for supported EV models can be obtained and used as the basis for route-feasibility estimation.
3. **External services will remain available during development and testing.**
   The project assumes that routing, geocoding, map, and AQI services can be accessed reliably enough for a pilot implementation.
4. **Charging-station and battery-swap data for the pilot area can be curated.**
   It is assumed that a sufficiently useful dataset of charging or swap locations in HCMC area can be collected and maintained..
5. **AQI data is suitable for area-based route comparison.**
   The project assumes that available AQI data can support zone-based pollution weighting, even though it may not represent exact street-level air quality.

6. **Users will access the system through supported platforms.**
   It is assumed that users will use either a modern web browser or an Android device with sufficient network connectivity for route planning and map display.
7. **Stakeholders will provide feedback in a timely manner.**
   It is assumed that the project supervisor and relevant reviewers will provide clarification and feedback early enough to support planning and implementation decisions.

Constraints

The project is subject to the following constraints:

1. **Time constraint**
   The project must be completed within the available capstone timeline. As a result, the system must focus on a limited MVP rather than a fully featured navigation product.
2. **Scope constraint**
   The MVP is limited to a **pilot area in Ho Chi Minh City** and mainly targets **electric motorbikes/scooters**. Broader geographic coverage and additional transport modes are outside the initial scope.
3. **Technology constraint**
   The project will rely on existing routing, map, geocoding, and AQI services instead of building a routing engine, telematics platform, or pollution-monitoring system from scratch.
4. **Data-quality constraint**
   Charging-station, battery-swap, and AQI data may be incomplete, outdated, or uneven in quality. This may limit the precision of route recommendations and pollution-aware scoring.
5. **Accuracy constraint**
   Battery-feasibility estimation is based on simplified range assumptions rather than direct connection to vehicle hardware. Therefore, the system can provide guidance and confidence levels, but not exact real-world battery predictions.
6. **Platform constraint**
   The MVP will support Android and web platforms only. Support on other platforms (e.g., IOS) is outside the current project scope.
7. **Privacy constraint**
   The project is designed to avoid central storage of personal trip data. While this reduces privacy concerns, it also limits features such as cross-device synchronisation, cloud backup, and server-side analytics on user behaviour.
8. **Resource and knowledge constraint**
   The project team must work within its available technical knowledge, development tools, and infrastructure budget. Some advanced features may need to be simplified or excluded if they require specialised expertise or paid services beyond the project's capacity.
9. **Requirement clarity constraint**
   Some requirements may evolve during the project as feasibility is tested and stakeholder feedback is received. This means the team must balance flexibility with the need to protect the MVP scope.
10. **Network and service dependency constraint**
    The system's performance depends partly on internet connectivity and the responsiveness

of third-party services. Weak connectivity or external-service issues may affect route generation and data loading during use.

2.6. Role and Responsibility

Role	Assigned Member	Skills	Responsibilities	WBS Task
Project Manager	Ma Thanh Long Mason	Project planning, coordination	Oversees project progress, schedules tasks, manages team communication	Team coordination, Listing/Validating product backlogs, Schedule management, Scope management, Progress tracking and evaluation
Front-end Developer	Do Hao Nhien, Tran Hoang Nguyen	HTML, CSS, JavaScript, UI design	Develops the website interface and ensures responsive design	Frontend Development (Desktop Interface), State Management, Web Environment Deployment, Frontend Production Build
Back-end Developer	Dao Ngoc Huy, Ma Thanh Long Mason	APIs, server-side programming, databases	Handles data processing, API integration, and backend logic	System Architecture (Design, Setup, DB), Backend Development (External API, REST API, Route Evaluation), Server/API Deployment
Mobile Application Developer	Ma Thanh Long Mason	Android development (Java/Kotlin), mobile UI	Develops the Android application and integrates APIs	Frontend Development (Mobile Interface), Mobile Application Distribution, Android WebView Wrapper Configuration, APK/Bundle Generation
UI/UX Designer	Shared role	Interface design, user experience principles	Designs layouts and ensures the system is user-friendly	User Interface (Design phase), Usability Testing (Mobile UI/UX, Desktop UI/UX)
Test/QA	Shared role	Testing, debugging	Tests the system and identifies bugs or usability issues	Issue Remediation, Integration Testing, Functional Testing, Testing Documentation, Post Deployment Verification

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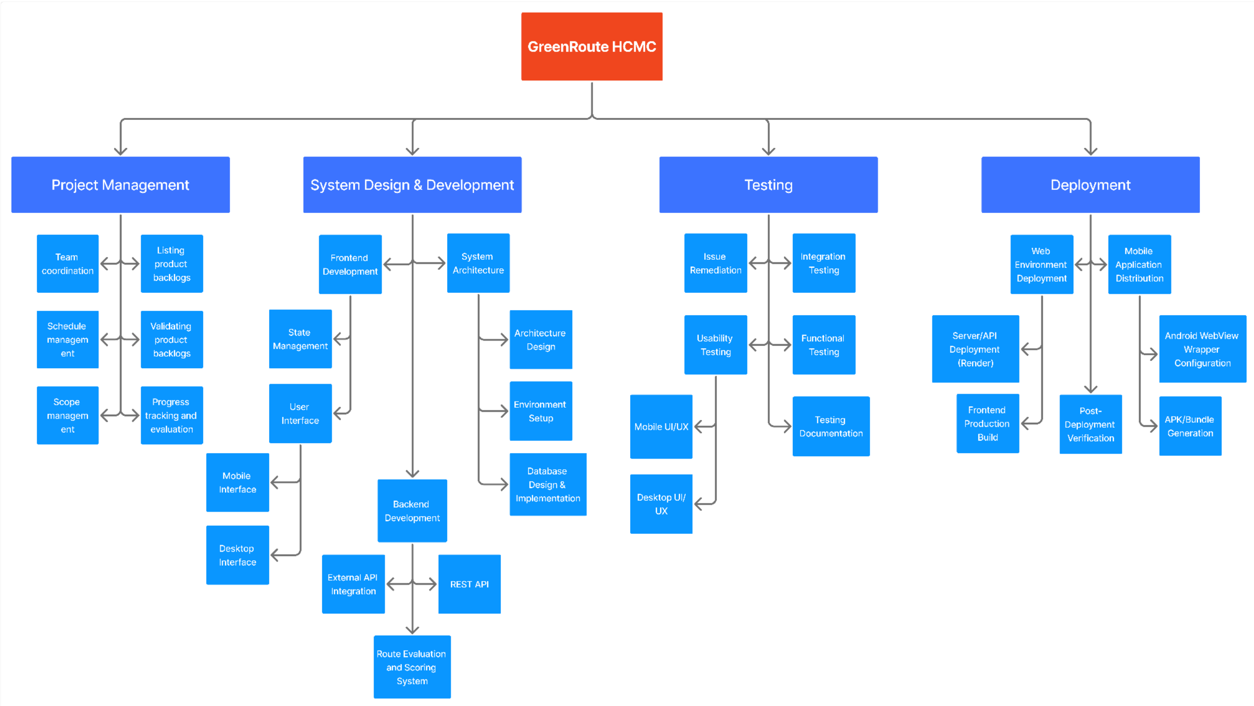
APPENDIX A: Software Resources Table

Software	**Purpose**
Visual Studio Code / Android Studio	Development environment for website and Android app.
GitHub	Version control and collaboration.
Web Browser (Chrome, Firefox, Edge)	Testing the web application.
Trello	Project management, task allocation, and progress tracking
ChatGPT, Gemini, Copilot	Provide assistance with documentation. Aid in establishing initial file/code infrastructure for the project.
JavaScript	Backend language
HTML + CSS	Frontend language
Java	Mobile development language
JSON + LocalStorage	Cache data for smoother user experience.
OpenStreetMap [5]	Open geographic database providing map data and points of interest.
Overpass API [8]	Query OpenStreetMap data to retrieve locations of charging stations and other map features.
NodeJS	Runtime environment used to run the backend server that processes external API requests for environmental data.
ExpressJS	Backend framework used to create REST endpoints for retrieving and serving pollution, weather, and location data to the web application.
Fetch	HTTP request interface used within the backend to retrieve data from third-party APIs such as air quality and weather services.
Leaflet [3]	JavaScript mapping library used to render interactive maps and overlays.
OSRM [4]	Routing API that calculates driving routes using OpenStreetMap road networks.
IQAir [6] + Nominatim [9]	Retrieve air quality index (AQI) data and match station locations to geographic coordinates.
Open-Meteo [7]	Weather Forecast API. For retrieving rain, wind, temperature, and

Software	Purpose
	humidity data.
WebView	Deployment of web application through the form of an android application.
Render	Online deployment of web application

APPENDIX B: Work Breakdown Structure



APPENDIX C: Project Schedule

The schedule below is a basic high-level walkthrough of the project's plan. Trello will be used to manage the project's schedule during the development period.

Trello Board Invitation: Link

Week 1 – Environment Setup & Prerequisites

- Lead: Backend Dev, Frontend Dev, Mobile Developer
- Duration: 1 week
- Focus: Project initialization and environment configuration
- Outcome: Functional development environments; backend and frontend base projects established

Small Milestone 1: Development environment fully operational

Week 2 – System Skeleton Development

- Lead: Backend Dev, Frontend Dev, Mobile Developer
- Duration: 1 week
- Focus: Core system structure using mock data and initial UI layouts
- Outcome: Basic end-to-end system (input → processing → display) using placeholder data

Small Milestone 2: Working system prototype with UI skeleton

Week 3 – Core Logic Implementation

- Lead: Backend Dev, Frontend Dev
- Duration: 1 week
- Focus: Implementation of scoring system for routing engine, pollution overlay, pollution-aware routing, and user interaction
- Outcome: Functional pathfinding system integrated with the user interface

Small Milestone 3: Fully functional routing system

Week 4 – Data Integration & Mid-Project Review

- Lead: Backend Dev, Project Manager
- Duration: 1 week
- Focus: Integration of pre-prepared APIs and system validation; presentation preparation
- Outcome: System operating with real data; mid-project evaluation completed

Large Milestone 1: Real data integration completed and validated

Week 5 – Environmental Overlays

- Lead: Backend Dev, Frontend Dev
- Duration: 1 week
- Focus: Implementation of weather and pollution overlays with interactive controls
- Outcome: Multi-layer environmental visualization integrated into the system

Small Milestone 4: Advanced map features completed

Week 6 – UI Completion & Mobile Integration

- Lead: Frontend Dev, Mobile Developer
- Duration: 1 week
- Focus: Finalization of web interface and integration into Android application
- Outcome: Fully usable system across web and mobile platforms

Small Milestone 5: Complete user interface and mobile compatibility achieved

Week 7 – Testing & Deployment ● Lead: Project Manager, Development Team ● Duration: 1 week ● Focus: System testing, bug fixing, and deployment via Render ● Outcome: Stable, deployed system with validated functionality **Large Milestone 2:** System successfully deployed
Week 8 – Project Closure & Documentation ● Lead: Project Manager, All Members ● Duration: 1 week ● Focus: Final presentation, documentation, and evaluation ● Outcome: Completed deliverables and formal project closure **Small Milestone 6:** Final project submission and sign-off

APPENDIX D: Risk Assessment

This project involves technical, estimation-related, requirement-related, and human-related risks that may affect its delivery, quality, or feasibility. To manage these risks consistently, the following assessment scale will be used:

Likelihood scale

- **High**: likely to occur during the project unless actively controlled
- **Medium**: possible to occur under normal project conditions
- **Low**: unlikely to occur, but still possible

Impact scale

- **1 – Negligible**: little to no effect on project outcome
- **2 – Minor**: small disruption, easily manageable
- **3 – Moderate**: noticeable impact on one or more features or milestones
- **4 – Major**: serious impact on scope, schedule, or system quality
- **5 – Critical**: threatens successful completion of the project

Risk Category	Risk Description	Likelihood	Impact	Mitigation
Human-related	Uneven workload distribution or unclear responsibility may lead to delays or incomplete features.	Medium	4	Clearly assign responsibilities for each module, track progress weekly, and ensure shared understanding of all core components.

Risk Category	Risk Description	Likelihood	Impact	Mitigation
Human-related	Temporary unavailability of a team member (e.g., illness, personal issues) may disrupt progress.	Medium	4	Maintain shared documentation and code access, and ensure that at least two members understand each critical module.
Human-related	Miscommunication within the team may cause integration issues or duplicated work.	Medium	3	Use regular check-ins, define clear task boundaries, and maintain consistent documentation of decisions and APIs.
Estimation-related	Underestimation of development effort for key modules such as routing logic, EV range calculation, or cross-platform integration.	Medium	5	Prioritise high-risk features early, break tasks into smaller milestones, and continuously adjust the plan based on progress.

Risk Category	Risk Description	Likelihood	Impact	Mitigation
Schedule-related	The project timeline may be too compressed for full implementation and testing of all planned features.	Medium	5	Focus strictly on MVP features first, delay non-essential enhancements, and allocate buffer time for testing and debugging. If the team still cannot deliver in time, we have to properly reflect on the reason why, and make an estimation of what was/is needed to deliver the missing features.
Requirements-related	Scope creep due to adding optional features (weather, offline mode, safety alerts) may affect completion of core features.	High	4	Freeze MVP scope early and only include enhancement features if core functionality is stable.
Technology-related	Integration issues between Android app, web app, backend API, and external services may delay development.	Medium	4	Define API contracts early, test integration incrementally, and avoid leaving integration to the final stage.

Risk Category	Risk Description	Likelihood	Impact	Mitigation
Technology-related	Inaccurate route guidance due to implementation logic errors or incorrect handling of routing results.	Medium	5	Validate route outputs with test cases, manually verify key routes, and simplify logic where necessary to improve reliability.
Data/implementation-related	Charging-station or map data used in the system may be incomplete or inconsistent within the pilot dataset.	Medium	4	Manually curate a reliable dataset for the demo area and verify important locations during testing.